|INCH-POUND|

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MILITARY SPECIFICATION

ELECTRO EROSION MACHINE, ELECTROLYTIC, CARBIDE TOOL BIT AND CHIP BREAKER GRINDER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers single and double end, floor mounted, motor driven, horizontal spindle, electrolytic grinding machines, suitable for grinding carbide chip breakers, inserts, and tool bits, with equipment and accessories specified herein.

1.2 Classification. The specification covers electrolytic grinding machines of the following styles, classes, and sizes. The style, class, and size for a specific procurement shall be as specified (see 3.5 and 6.2.1).

Class I - Single End, Tool Bit Grinder

Style A - Single Table

Size	Wheel diameter (inches)
6	6
7	7
10	10

Style B - Dual Table

Size	Wheel diameter (inches)
10	10
14	14

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, 2163 Airways Boulevard, Memphis, Tennessee 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Class II - Single End, Chip Breaker Grinder

Size	Wheel diameter (inches)
6	6
10	10

Class III - Double End, Combination Tool Bit and Chip Breaker Grinder

Size	Wheel diameter (inches)
6	6
7	7

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interferences.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

(Unless otherwise indicated, copies of Federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this

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document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

U. S. DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402-0001.)

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN GEAR MANUFACTURERS' ASSOCIATION (AGMA)

AGMA 218.01 - Standard for Rating the Pitting Resistance and Bending Strength of Spur and Helical Involute Gear Teeth.

AGMA 390.03a - Gear Classification, Materials and Measuring Methods for Bevel, Hypoid, Fine Pitch Wormgearing and Racks Only as Unassembled Gears.

AGMA 2000-A88 - Gear Classification and Inspection Handbook.

(Application for copies should be addressed to the American Gear Manufacturers' Association, Standards Department, 1500 King Street, Suite 201, Alexandria, VA 22314-2717.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B5.1 - T-Slots, Their Bolts, Nuts, Tongues, and Cutters. (DOD adopted)

ANSI B74.2 - Specifications for Shapes and Sizes of Grinding Wheels, and Shapes, Sizes, and Identifications of Mounted Wheels. (DOD adopted)

ANSI/ISO 54 - Cylindrical Gears For General Engineering and for Heavy Engineering, Modules and Diametral Pitches of.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018-3363.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

JOINT INDUSTRIAL COUNCIL (JIC)

JIC H-1 - Hydraulic Standards for Industrial Equipment and General Purpose Machine Tools.

(Application for copies should be addressed to the Joint Industrial Council, c/o The National Machine Tool Builders' Association, 7901 Westpark Drive, McLean, VA 22101-4269.)

NATIONAL ELECTRICAL MANUFACTURER'S ASSOCIATION (NEMA)

NEMA MG 1 - Motors and Generators. (DOD adopted)

(Application for copies should be addressed to the National Electrical Manufacturers' Association, 2101 L Street, N.W., Washington, DC 20037-1580.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery. (DOD adopted)

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), one complete machine shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

3.2 Design. The electro erosion machine shall be new (not a prototype) and one of the manufacturer's current production models capable of operations in accordance with the requirements herein. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment replacement, and repair.

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3.2.1 Measurement systems. Unless otherwise specified (see 6.2), both the U.S. Customary System of Units (US) and the International System of Units (SI) may be used in the design and construction of the electro erosion machine in this specification, all measurements, dimensions, sizes, and capacities are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.2.1.1 Measuring and indicating device calibrations. Measuring and indicating devices shall be graduated in US, SI, or dual (US and SI) units as specified by the procuring activity (see 6.2). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system.

3.2.1.2 Dials. All dials used to indicate worktable and cross-slide position shall be graduated in increments not larger than 0.001 inch. Each fifth shall be larger than the others and each tenth graduation shall be numbered. Dial and handwheel diameters shall be permanently and legibly engraved or etched on a nonglare background with graduations that can be read from the operator's normal position.

3.2.1.3 Digital readout system. Unless otherwise specified (see 6.2), a digital readout system shall be installed on the X and Y (axis, axes) of travel. The system shall consist of a position feedback device, an electronic display for each axis, and all related components, mounting hardware, and electrical equipment necessary to ensure complete operation of the system. The system shall provide a readily visible digital display over the full working range of each axis on which installed. The system resolution shall be 0.0001 inch. The system accuracy shall be equal to the system resolution, verified upon installation and calibration of each system on the machine. The repeatability shall be 0.0001 inch, measured over the full travel on each axis. Special requirements for the system components shall be as specified (see 6.2). The display shall indicate axis location in both the inch-pound and metric units. An input power source fluctuation of $\pm$ 10 percent shall not adversely affect the system operation.

3.2.1.4 Position feedback device. A position feedback device including all related components, hardware, and electrical equipment shall be provided for each axis having a readout system. The feedback device shall be of the independent type with a resolution equal to the resolution of the digital readout system. Each feedback device shall be installed with a cover or guard to prevent physical damage to any portion of the feedback device. The connecting cable from the position feedback device shall be electronically shielded to protect against electronic interference and shall be armored to protect against contamination and damage. The device shall provide axis travel feedback over the entire range of axis travel for each specified axis. Installation of the position feedback device shall not limit machine travel on any axis nor eliminate manual machine stops. Provisions shall be made to prevent damage to the feedback device in the event of overtravel. The feedback unit and related components shall be protected from contamination. Position feedback devices shall be easily and quickly interchangeable between like machines procured under the same contract.

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3.2.1.5 Electronic display console. The electronic display console for the digital readout system shall contain a universal digital display with individual illuminated digits of not less than 0.43 inch in height. Digits shall be high efficiency displays for high light ambient conditions with wide viewing angle and shall be readable at a distance of not less than 16 feet. The console shall display at least six digits per axis simultaneously. The display console shall also provide keypad entry, memory, absolute/incremental counting, data preset, power failure indicator, inch/metric conversion, and absolute zero recall. The electronic display console shall be installed so as to be readily accessible and easily read from the operator's normal work station. The console shall be mounted to withstand shock and vibration generated by the machine. Each connecting cable shall be armored and shall be securely fastened to the structure of the machine. Location of installed cables shall prevent any interference with normal machine operation. Electronic display consoles shall be easily and quickly interchangeable between like machines procured under the same contract.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials to the maximum extent possible provided such materials will not jeopardize the intended use, performance, or design life of the machine. Reclaimed materials shall have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used, rebuilt, or remanufactured end products are allowed under this specification.

3.2.3 Energy efficiency. The electro erosion and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency in accordance with by the developments available within the industry.

3.2.4 Controls. All mechanical, hydraulic, and pneumatic operating controls shall be located convenient to the operator's work station(s).

3.2.5 Safety and health requirements. All parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall conform to all requirements of OSHA 29 CFR 1910 and ANSI B11.9. If a conflict arises between 29 CFR 1910 and ANSI B11.9, 29 CFR 1910 shall apply. Additional safety and health requirements shall be as specified (see 6.2). Covers, guards, or other safety devices normally furnished as standard on the manufacturer's commercial machine supplied to the commercial market shall be provided for the point of operation and all other parts of the machine that present safety hazards. If point-of-operation guarding is influenced by workpiece size and configuration, the procuring activity shall provide part design data to the supplier of the machine (see 6.2). The supplier of the machine shall provide the special guarding which shall meet OSHA 29 CFR 1910.

3.2.6 Mercury restriction. The electro erosion machine shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the electro erosion machine.

3.2.8 Environmental protection. During the manufacture, operation, service, transportation, or storage of the machine and its components, materials hazardous to the ecological system as prohibited by applicable Federal, state,

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or local statutes in effect on the date of the contract shall not be used or emitted.

3.2.9 Lubrication. Means shall be provided to ensure adequate lubrication for all moving parts. Recirculating lubrication systems shall include a cleanable or replaceable filter. Means for determining oil level in each reservoir shall be by an oil level sight gauge, a low oil level warning light, or similar device and shall be in clear view of the operator's normal work station. An automatic lubrication system shall be provided for all ways and axis screws. Each lubricant reservoir shall have at least a 24-hour capacity and means for determining fluid level. All oil holes, grease fittings, and filler caps shall be readily accessible.

3.2.10 Interchangeability. To provide for replacement of worn parts, all parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or machine. Replacement parts shall be domestically available.

3.3 Construction. The electro erosion machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all operation to its maximum rating and capacity without permanent distortion or failure.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Fastening devices. All fasteners shall be installed to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. Any materials used for such operations shall be thoroughly removed from the part(s) upon completion of the operations.

3.3.5 Painting. Unless otherwise specified (see 6.2), the paint color and application shall be in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts used on the electro erosion machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Drive train gears. All gears of spindle and axis drive trains shall conform to or exceed all provisions of AGMA 390.03a, AGMA 218.01, and AGMA 2000-A88 for the inch-pound system of measurement or ANSI/ISO 54 for the metric system of measurement. The gears shall be manufactured of a suitable ferrous material. The gears shall be of proper width and size to transmit full-rated

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torque and horsepower throughout the speed ranges without failure for the expected service life of the machine. Gears in the drive train shall be hardened and ground steel. Working surface hardness shall be not less than Rockwell C-48.

3.3.8 Electromagnetic interference control. When specified (see 6.2), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystem(s) class and the emission and susceptibility requirements shall be as specified.

3.4 Components. The machine shall consist of not less than a base, spindle, grinding wheel(s), worktable, workholding vise, and electrolyte feed system.

3.4.1 Base. The base shall be a floor mounted casting or fabricated steel weldment. The structure shall be internally reinforced, stress relieved, and normalized to minimize distortion and deflection. If drive motors, electrolyte reservoir, or other internal mechanisms are mounted in the base, doors or readily removable covers shall be provided for servicing the components.

3.4.2 Spindle. The spindles shall be ground, machine finished dynamically balanced, and hardened. The spindles shall rotate on preloaded ball or roller bearings, have power pickup for electrolytic grinding, and electrically insulated from other parts of the grinder. The spindles shall be single end for class I and class II machines, and double end for class III machines. The spindle end (s) shall be designed to accommodate grinding wheels conforming to 3.4.4 through 3.4.4.3.

3.4.3 Spindle drive. The spindle drive shall be either direct, as an integral part of the motor shaft, or by belt from the motor to the spindle. When belt drive is used, means shall be provided for adjustment of belt tension. Motor horsepower shall meet the requirements of table I.

3.4.4 Grinding wheels. Unless otherwise specified (see 6.2), one grinding wheel shall be furnished with each class I and class II machine and two grinding wheels (one for each end of the double-end spindle) on class III machines. Unless otherwise specified (see 6.2), the grinding wheels shall be of the diamond impregnated metal bonded type having a minimum material. The wheels shall be designed specifically for electrolytic grinding of cemented carbide cutting tools and shall meet the performance requirements stated herein. Aluminum shall not be used for the hub material or the matrix material of the wheels. When mounted on the machine spindle, the wheels shall run true within an accuracy of 0.0005 inch runout (total indicator reading) in both the radial and lateral direction measured at the perimeter of the wheel. In addition, the grinding wheels shall conform to the requirements of 3.4.4.1 through 3.4.4.3, as applicable for the class machine specified.

3.4.4.1 Grinding wheel for class I machines. The grinding wheel for class I machines shall be the plate mounted, cup type. The wheel shall be suitable for grinding carbide tools and inserts to specific geometrical shapes. The wheel shall conform to the shape specified in ANSI/B74.3.

3.4.4.2 Grinding wheel for class II machines. The grinding wheel for class II machines shall be the chip breaker type. The wheel shall be suitable for grinding chip breaker grooves, or steps, on the top face of carbide cutting tools or inserts.

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3.4.4.3 Grinding wheels for class III machines. The grinding wheels for class III machines shall be the cup type for the tool bit grinding end of the machine, and the chip breaker end of the machine.

3.4.5 Work tables.

3.4.5.1 Worktable for class I machines. The worktable for class I, style A machines shall be mounted on ways and shall feed into the grinding wheel. Both a manual and a powered table infeed mechanisms shall be provided having dials to indicate worktable position. The power feed mechanism may be either electric or hydraulic or a combinations of these methods. Means shall be provided to stop the feed motion at a preadjusted position or after a preadjusted time interval. A table lock shall be provided to lock the table at any position for setup purposes. The table shall tilt on an axis parallel to the plane of the grinding wheel face to allow grinding of relief angles. A scale or dial graduated in degrees shall be provided for indicating the table tilt angle and a clamping device shall be furnished to securely clamp the table at any angle within the tilt range specified in table I. The table shall have a T-slot conforming to ANSI B5.1 for aligning and bolting a swivel type workholding vise, conforming to 3.4.6.1, to the table surface. The table top shall have replaceable wear plates. Style B machines shall be similar to and shall have all the design characteristics of style A machines, except style B machines shall have dual tables and dual controls.

3.4.5.2 Worktable for class II machines. The worktable for class II machines shall be mounted on ways and shall reciprocate parallel to the grinding wheel face. The table shall be reciprocate manually by lever or handwheel. Adjustable stops shall be provided for regulating the stroke length. Table traverse shall conform to table I. The table shall have ways to support a cross slide. The cross slide shall move parallel to the axis of the grinding wheel. The cross shall have means for alignment and securely holding a swivel vise that conforms to 3.4.6.2. The table shall also have an elevating mechanism to allow the table, cross slide, vise, and tool to be elevated and lowered. The cross slide feed screw and the elevating feed screw shall have dials to indicate worktable and cross slide position.

3.4.5.3 Worktables for class III machines. Worktables for class III machines shall conform to 3.4.5.1 for the cupwheel side of the machine and 3.4.5.2 for the chip breaker side of the machine.

3.4.6 Workholding vise.

3.4.6.1 Workholding vise for class I machines. The class I machine shall have a swivel vise that mounts on the worktable, holds the tool being ground, and allows the tool to be swiveled to the wheel face. The vise shall swivel not less than 180 degrees, shall be provided with a dial graduated in one degree increments, and shall have means for clamping at any swivel angle. The vise shall have a tongue or key to align the vise to the axis of the table and means for securely bolting the vise to the table. The vise capacity shall be not less than 1-1/2 inch. Special types of workholder vises, when required, shall be as specified (see 6.2).

3.4.6.2 Workholding vise for class II machines. The class II machines shall have a double tilting-swiveling vise mounted on the cross slide to hold

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the tool being ground, and allow the tool to be tilted and swiveled in relation to the chip breaker wheel. The vise shall swivel through an angle not less than 180 degrees and shall tilt, or rock, about two axes of rotation. One tilt motion shall be perpendicular to the axis of the wheel, and the other tilt motion shall be parallel to the axis of the wheel. A dial graduated in increments not larger than one degree shall be provided indicate the angles of tilt and swivel. Means shall be provided to securely clamp the vise at any tilt or swivel angle. The vise capacity shall be not less than 2-1/2 inches.

3.4.6.3 Workholding vise for class III machine. The workholding vises fore class III machines shall conform to 3.4.6.1 for the cupwheel side of the machine and 3.4.6.2 for the chip breaker side of the machine.

3.4.7 Oscillating mechanism. Class I and class III machines shall be provided with an oscillating mechanism to oscillate either the wheel or the worktable(s). The stroke length and frequency of the oscillating part shall conform to the requirements of table I. Means shall be provided to lock the oscillating part when oscillation is not required. A safety device shall be provided to prevent engaging the oscillating mechanism when the oscillating wheel or table is locked.

3.4.8 Electrolyte feed system. The machine shall be equipped with a complete electrolyte feed system. The system shall include piping, mist collection system, storage tank, pump, and filter. Troughs or trays on the table top for returning spend electrolyte to the storage tank shall be provided. Recirculated electrolyte shall be filtered. The electrolyte reservoir shall have a capacity of not less than 20 gallons.

3.4.9 Power supply. The grinding machine shall be equipped with a direct current power supply with output terminals connected in such a manner that the workpiece is anodic and the grinding wheel cathodic. Means shall be provided to reverse the polarity for dressing the grinding wheel. The power supply unit shall have a continuous output current rating of not less than 150, 300, 600, 800, 1500, or 3000 amperes as specified (see 6.2).

3.4.10 Hydraulic system. The hydraulic, when provided, system shall meet JIC H-1. The hydraulic system shall be complete, including all pumps, valves, piping, cylinders, and pressure controls. Overpressure protection shall be provided in the high pressure line to prevent damage to components. A filter system shall be provided to ensure delivery of clean fluid. The hydraulic reservoir shall have means for determining fluid level, a provision for draining, and a cleanout access plate if the reservoir is not removable.

3.4.11 Electrical system. The electrical system shall conform to NFPA 79. Unless otherwise specified (see 6.2), each machine shall draw all of its electrical power on a single 230/460-volts, 3-phase, 60-hertz (Hz) circuit having a fused safety disconnect switch or circuit breaker. The machine shall be initially wired for operation on 460 volts.

3.4.11.1 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Unless otherwise specified (see 6.2), alternating-current (AC) motors shall be designed to operate on 60-Hz. The motor enclosure shall meet the requirements for a drip-proof enclosure. All motors shall conform to the requirements of NEMA MG-1.

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3.4.11.2 Worklight. When specified (see 6.2), a worklight shall be provided. The light shall have a protective shield, an on-off switch, and shall be adjustable to reflect maximum light on the work area. If a built-in lighting system is provided, the adjustable light described above shall not be required.

3.4.11.3 Hour meter. When specified (see 6.2), the machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 99,999 hours in increments of not greater than 1 hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine.

3.5 Size and capacity. Unless otherwise specified (see 6.2), the size and capacity of the machine shall meet the requirements of table I for the machine specified. Machine capacity shall be not less than the stated requirements of the size ordered. When a range is given, the required performance shall be from the stated minimum or less to the stated maximum or greater.

3.6 Performance. The machine shall be capable of electrolytically grinding carbide tool bits, carbide inserts or slugs, and chip breakers to applicable prescribed dimensions and rates specified in 4.8.

3.6.1 Addition performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2).

3.7 Standard equipment. All standard equipment normally provided with the manufacturer's commercial machine shall be furnished.

3.7.1 Optional equipment. Optional equipment shall be furnished as specified and full described (see 6.2). The optional equipment provided shall be functional without requiring modification of the equipment or machine.

3.8 Repair parts. Required repair parts shall be furnished as specified (see 6.2).

3.9 Marking on instruments, control panels, charts, and plates. All words on instruments, control panels, charts, and plates shall be in the English language. Characters shall be permanently marked in boldface on a contrasting background. All plates shall be corrosion resistant.

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3.9.1 Lubrication plate. When specified (see 6.2), a lubrication plate shall be attached to each machine. Unless otherwise specified (see 6.2), the information provided on the plate shall be as listed below.

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity
Federal or military specification for each lubricant
Size and type of each lubricant filter (if applicable)

3.9.2 Nameplate. A nameplate shall be securely attached to each machine. Unless otherwise specified (see 6.2), the nameplate shall contain the information listed below.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.10 Workmanship. Workmanship of the machine shall meet all requirements specified herein and shall be of a quality equal to that prevailing among manufacturers producing equipment of the type covered by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements, examinations, and tests specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements. Unless otherwise specified in the contract or purchase order, all inspections shall be performed within the limits of the continental U.S.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submissions of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be examinations shall be performed in an indoor facility with ambient conditions of 55° F to 100° F and 10 percent to 95 percent relative humidity.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. The machine shall pass the examination and all tests to be accepted.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the examination in 4.6, the tests in 4.7, and the inspection in 4.8. The machine shall pass the examination, all tests, and the inspection to be accepted.

4.6 Examination. The machine shall be examined to determine compliance with all requirements of this specification.

4.7 Tests.

4.7.1 Operational test. Each machine shall be operated at no load for not less than 1 hour(s). Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the trial period.

4.7.2 Performance test. Each grinding machine shall be subjected to tests as follows:

a. Class I machines are required to perform only those tests in 4.7.2.1 and 4.7.2.2.

b. Class II machines are required to perform only those tests in 4.7.2.3 and 4.7.2.4.

c. Class III machines are required to perform all tests in 4.7.2.1 through 4.7.2.4.

4.7.2.1 Solid slug plung grinding. A solid carbide slug of each of the grades shown below, 1/2 inch by 1/2 inch in cross-section, shall be plunge ground on the cupwheel at the current shown with a feed pressure of 20 pounds total force. Stock removal rates shall be not less than those shown.

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Characteristic	Requirement	
Carbide Manufacturer's Number	C-2, General Purpose cast iron and non-ferrous materials	C-5 Roughing cuts, steel
Linear removal rate, IPM	0.016	0.040
Amperage, plus or minus 5	85 amps	175 amps

4.7.2.2 Single point grinding. A single point carbide tool with carbide tip, dimensions of 1/2 inch wide by 1/4 inch thick by 7/8 inch long, shall be ground electrolytically on the cupwheel. The part of the tip to be ground, amount of stock to be removed, and the maximum grinding time shall be as follows:

Position	Stock Removal	Time in Seconds
Grind carbide tip on side	1/16	50
Grind carbide tip on top	1/64	45

4.7.2.3 Solid slug grooving. A solid carbide slug, 1/2 inch by 1/2 inch in cross-section, shall be ground on the chip breaker wheel, grinding a groove 1/4 inch wide by 0.50 inch deep by 1/2 inch long. Minimum average shall be 120 amperes and grinding time shall not exceed 30 seconds.

4.7.2.4 Single point grooving. A chip breaker groove on a single point tool shall be ground electrolytically. The groove shall be 1/8 inch wide by 1 inch long by 0.025 inch deep. Total grinding time shall be not more than 20.

4.7.3 Electromagnetic interference control test. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.8 using the procedures given in MIL-STD-462.

4.8 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-18058.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Grinding machines covered by this specification are intended for electrolytic carbide tool bit and chip breaker grinding operations normally required in general machine shop practice.

6.2 Acquisition requirements. Acquisition documents should specify the following:

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a. Title, number, and date of this specification.

b. Style, class, and size required (see 1.2).

c. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1. and 2.2).

d. First article, if required (see 3.1).

e. System of units for measuring, if different (see 3.2.1).

f. System of units for measuring and indicating devices (see 3.2.1.1).

g. Digital readout system, if not required (see 3.2.1.3).

h. Specify any special digital readout system features, if required (see 3.2.1.3).

i. Additional safety and health requirements, if required (see 3.2.5).

j. Painting, if different (see 3.3.5).

k. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.8).

l. Grinding wheel, if not required (see 3.4.4).

m. If grinding wheel is other than diamond impregnated specify type (see 3.4.4).

n. Special workholder vises, if required (see 3.4.6.1).

o. Power supply output current rating, as specified (see 3.4.9).

p. Electrical system, if different (see 3.4.11).

q. Motor enclosure, if different (see 3.4.11.1).

r. Worklight, if not required (see 3.4.11.2).

s. Hour meter, if required (see 3.4.11.3).

t. Size and capacity, if different (see 3.5).

u. Additional performance features, if required (see 3.6.1).

v. Specify and fully describe optional equipment, if required (see 3.7.1).

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w. Specify applicable repair parts, if required (see 3.8).

x. Lubrication plate, if different (see 3.9.1).

y. Nameplate, if different (see 3.9.2).

z. Inspection conditions, if different (see 4.3).

aa. First article inspection, if different (see 4.4).

bb. Quality conformance inspection, if different (see 4.5).

cc. Performance test, if different (see 4.7.2).

dd. If preservation, packing, and marking in accordance with MIL-M-18058 is required, specify level required and specification number (see 5.1).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
.	.	.	.
.	.	.	.

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.3.1 Technical data. It should be specified in the "Remarks" section of the DD Form 1424 that all technical data ordered shall be written in the English language.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

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6.5 Definitions.

6.5.1 Inch-pound units. Inch-pound units are a system of measures based on the yard and pound commonly used in the United States of America and defined by the National Institute of Standards and Technology. Inch-pound units having the same names in other countries may differ in magnitude.

6.5.2 Metric units. Metric units are a system of basic measures defined by the International System of Units based on "Le Systeme International D'Unites (SI)," of the International Bureau of Weights and Measures. These units are described in ASTM E 380 and IEEE 268 (MIL-STD-961).

6.6 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.7 Safety and health requirements. OSHA 29 CFR 1910 limits only the total hazard level (noise, radiation, electromagnetic emissions, noxious vapors, air contaminants, and heat) of the environment in which a machine will operate. It does not limit the hazard level of individual machines in an operating environment. The procuring activity should analyze the existing hazard level in the proposed operating environment and specify additional requirements necessary to integrate this new machine into its future environment.

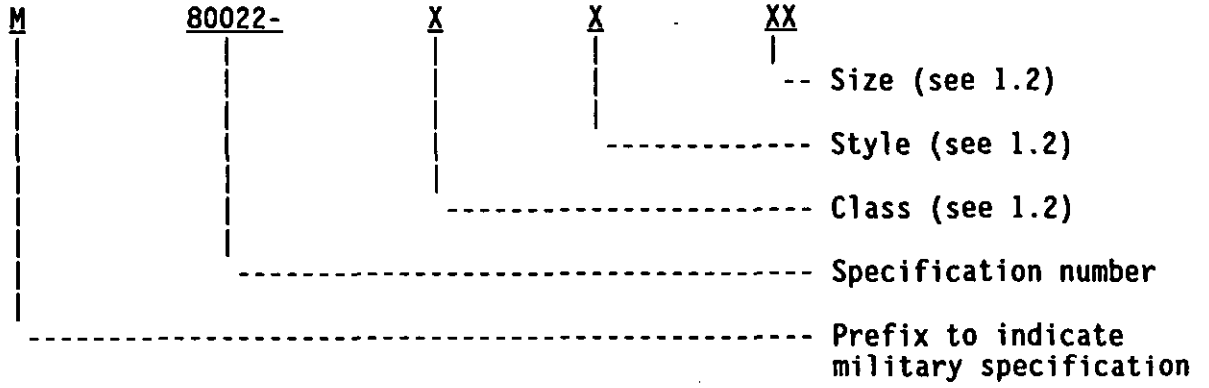
6.8 Training. Training required by the procuring activity should be provided as specified in the contract.

6.9 Warranty. Warranty requirements should be as specified by the procuring activity in the contract.

6.10 Inspection location. The contractor should identify in his response to the solicitation the location where inspection and tests are to be performed.

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6.11 Part or Identifying Number (PIN). The PIN to be used for machines acquired to this specification are created as follows:



(Note: Arabic numbers shall be used in the PIN.)

6.12 Subject term (keyword) listing.

Base
Grinding wheels
Spindle
Spindle drive
Workholding vise
Worktables

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TABLE I. Sizes and capacities

Characteristics	Requirement								
	Class I					Class II		Class III	
	Style A			Style B					
	Size 6	Size 7	Size 10	Size 10	Size 14	Size 6	Size 10	Size 6	Size 7
Spindle:									
Spindle speed, RPM	3500	3000	2150	2150	1650	3500	2150	3500	3000
Spindle motor, HP	2	2	5	5	5	2	2	2	2
Cup Wheel End:									
Work table, number	1	1	1	2 each	2 each	1	1	2	2
Width by length, inches	10 by 20	10 by 20	10 by 20	10 by 15 each table	10 by 10 each table	--	--	10 by 20	10 by 20
Work table tilt, Degrees from horizontal	25 down to 10 up	25 down to 10 up	25 down to 10 up	25 down to 10 up	22 down to 10 up	-- --	-- --	25 down to 10 up	25 down to 10 up
Oscillation stroke, inches	0 to 1-1/2	0 to 1-1/2	0 to 1-1/2	0 to 1-1/2	0 to 1-1/2	--	--	0 to 1-1/2	0 to 1-1/2
Frequency, oscillations per minute	60	60	60	60	50	--	--	60	60
Wheel size, diameter by thickness, inches	6 by 3/4	7 by 1	10 by 1	10 by 1	14 by 1	--	--	6 by 3/4	7 by 1
Wheel bore, inches	1-1/4	1-1/4	2	2	2	--	--	1-1/4	1-1/4
Wheel speed, surface feet per minute	5500	5500	5500	5500	5500	--	--	5500	5500
Chip Breaker Side:									
Table traverse, inches	----	----	----	----	----	8	8	7	7
Wheel size, diameter by thickness, inches	----	----	----	----	----	6 by 1/2	10 by 1/2	6 by 1/2	7 by 12
Wheel bore, inches	----	----	----	----	----	1-1/4	1-1/4	1-1/4	1-1/4
Wheel speed, surface feet per minute	----	----	----	----	----	5500	5500	5500	5500

NOTE: Requirements are not less than those shown. Where a range is shown, the required performance is from the stated minimum or less to the stated maximum or greater.

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CONCLUDING MATERIAL

Custodians:
Army - AL
Navy - SH
Air Force – 99

Preparing Activity:
DLA - IP

(Project 3410-0040)

Review Activities:
Air Force – 84

User Activities:
Navy – OS, AS, MC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-E-80022C	Electro Erosion Machine, Electrolytic, Carbide Tool Bit and Chip Breaker Grinder

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: ____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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